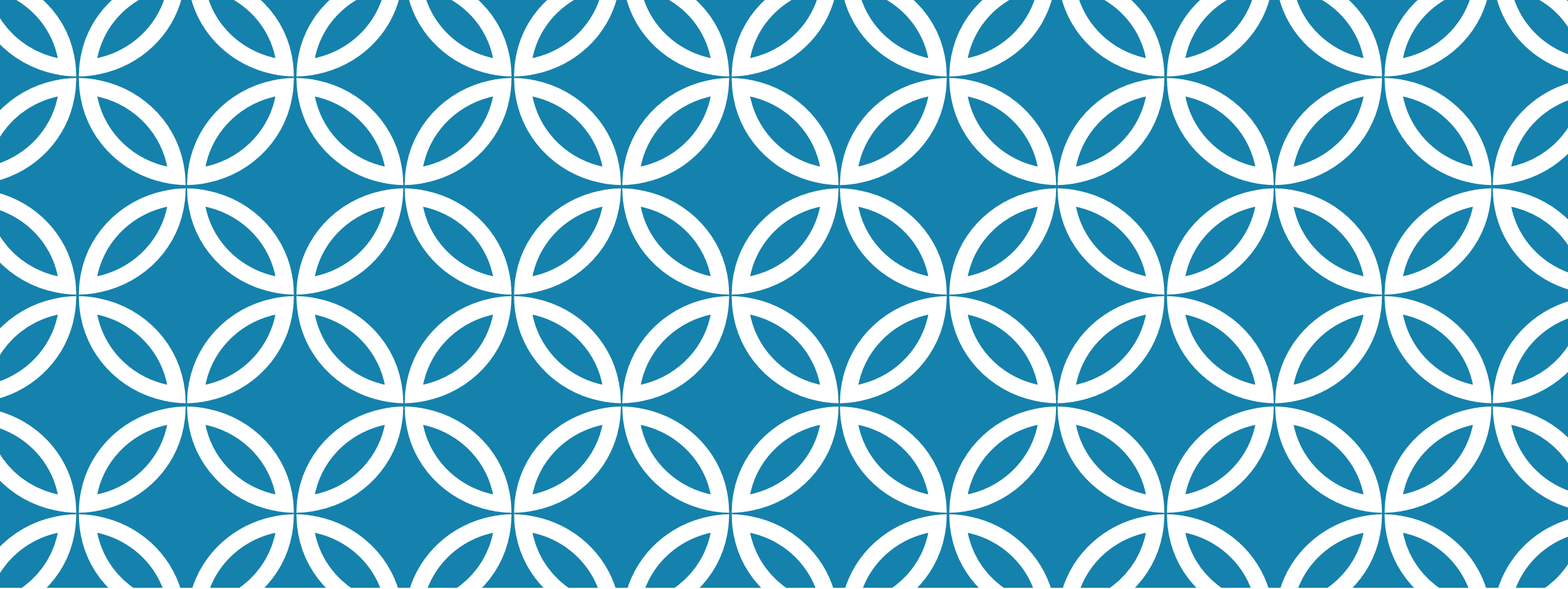




FROM ZERO TO CLOUD: BEGINNER'S GUIDE TO AWS EC2 AND COMPUTE SERVICES

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technologies

AGENDA

Amazon EC2 basics

- A scenario
- Amazon EC2 instance families
- Decoding instance names
- Amazon EC2 instance pricing
- Tools to right size workload
- Advantages of updating Instance type

AWS Compute Services

Demo

QnA

AMAZON EC2 BASICS

Compute services are the backbone

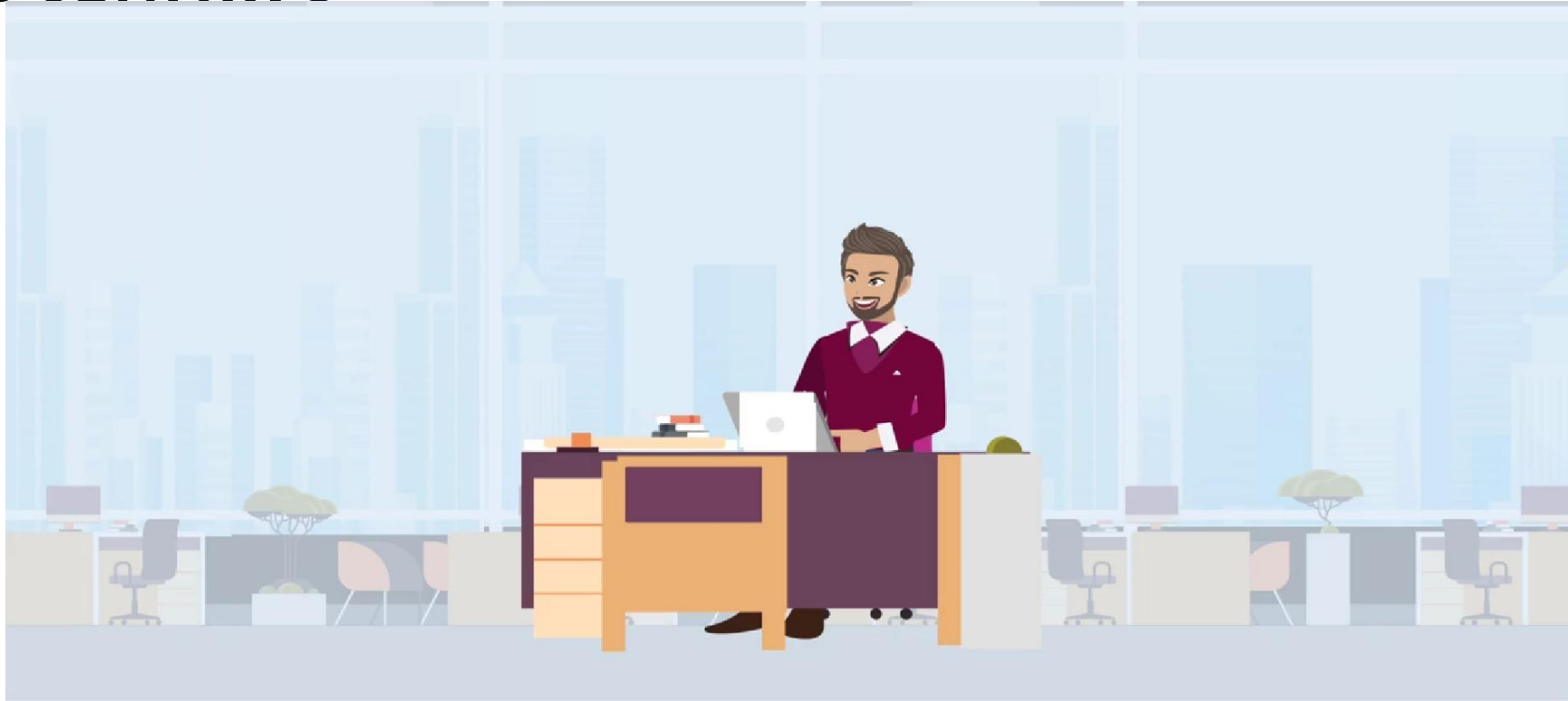
Amazon EC2 is the foundation

Different families of EC2 instances

Balancing cost and performance when selecting an EC2 inst. Type

Advantages with new instance types

A SCENARIO



Mr. John looking to move his on-premises workloads into the AWS cloud

Image courtesy: AWS skill builder

WHERE TO BEGIN?



EC2 instance families and difference between them?

Image courtesy: AWS skill builder



Ms. Sofia a Technical project lead and mentor

Image courtesy: AWS skill builder

AMAZON EC2 INSTANCE FAMILIES

850+ instance types

Latest processor, storage, networking and operating system to match the workload

Each instance name references the underlying function
and hardware resources available to that instance

INSTANCE TYPES

General purpose

General purpose instances provide a balance of compute, memory and networking resources, and can be used for many workloads. These instances are good for applications such as web servers, code repositories, and small-to-medium...

Compute optimized

Compute optimized instances are ideal for compute bound applications that benefit from high-performance processors. Some examples of workloads for compute instances are batch processing, media transcoding, and dedicated...

Memory optimized

Memory optimized instances are designed to deliver fast performance for workloads that process large data sets in memory. For example, these instances are good for in-memory databases, data analytics, and enterprise...

Accelerated computing

Accelerated computing instances use hardware accelerators, or co-processors, to perform functions more efficiently. For example, they can perform floating point number calculations, graphics processing, or data pattern matching.

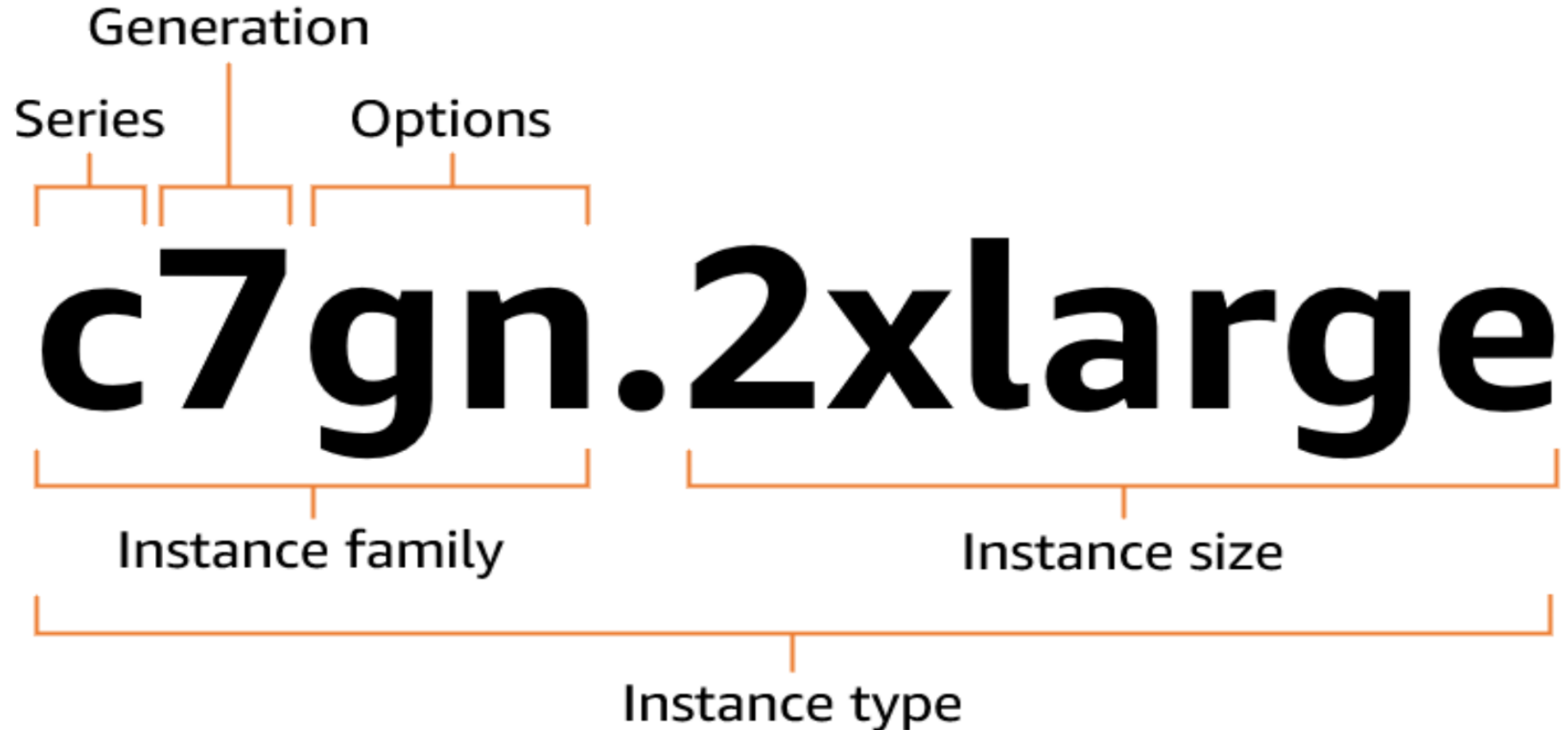
Storage optimized

Storage optimized instances deliver millions of low-latency, random I/O operations per second to applications. They're designed for workloads that require high, sequential read and write access to very large data sets on local storage. For exampl...

HPC optimized

High-performance computing (HPC) instances offer the best price performance for running HPC workloads at scale. HPC instances are ideal for applications that benefit from high-performance processors such as complex simulations, deep...

DECODING INSTANCE NAMES



AMAZON EC2 INSTANCE PRICING

On-demand instances

Spot instances

Saving plans

Reserved instances

Dedicated hosts

TOOLS TO RIGHT SIZE WORKLOAD

AWS Compute Optimizer

Pricing calculator

Cost explorer

Heath Dashboard



ADVANTAGES OF UPDATING INSTANCE TYPE

Lower pricing

Faster performance

Benefits of the new generation

AWS COMPUTE SERVICES

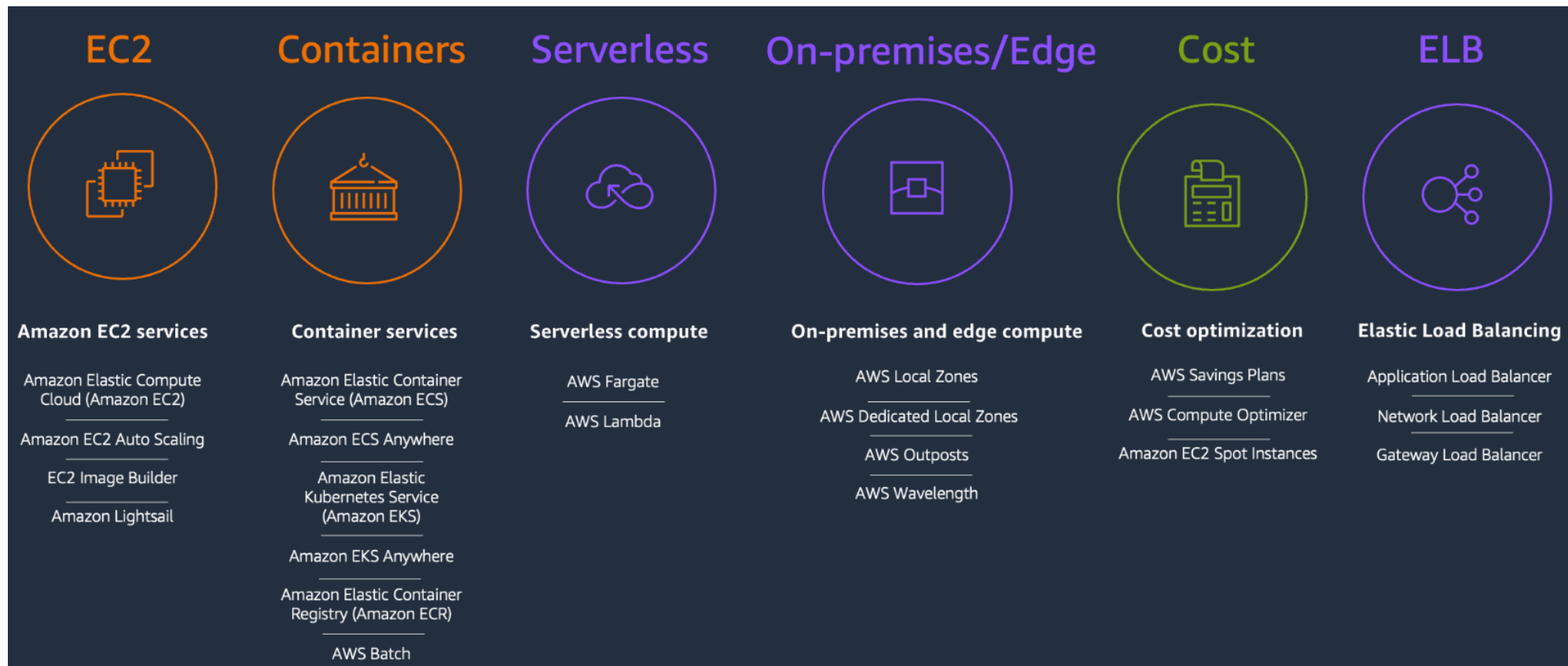


Image courtesy: AWS portal

DEMO

- Launch an EC2 instance Linux instance and connect with it
- Launch an EC2 Windows instance and connect with it
- Launch EC2 instances using user data
- Create an elastic load balancer(Application Load balancer)
- Create auto scaling group



QNA

REFERENCES

<https://aws.amazon.com/ec2/instance-types/>

<https://aws.amazon.com/ec2/pricing/>

<https://aws.amazon.com/ec2/faqs/#topic-0>

<https://docs.aws.amazon.com/ec2/latest/instancetypes/instance-type-names.html>

LETS CONNECT



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